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Path compromise and defenses for long-lived mixnet communication

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Outline

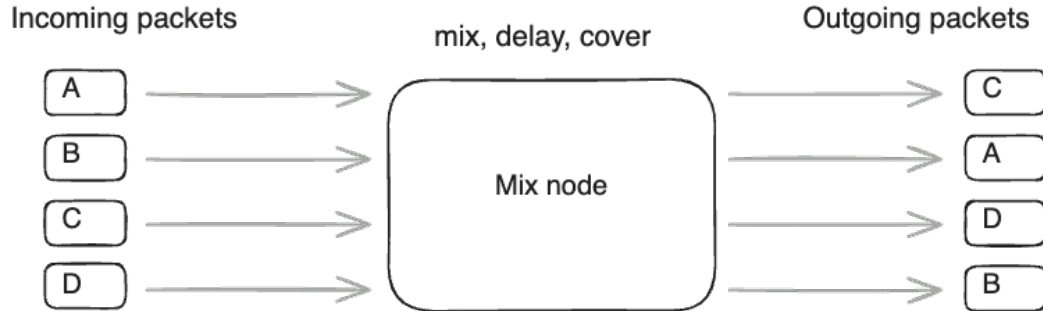
- **Brief background/motivation**
- **What is the path compromise problem?**
- **Existing solutions/defences**
- **Do they apply to the mix setting?**
- **How do we simulate it?**
- **Can we do better?**
- **Evaluation/results**

Mix protocol

Goal: a malicious observer cannot determine or correlate the input and output of the mix.

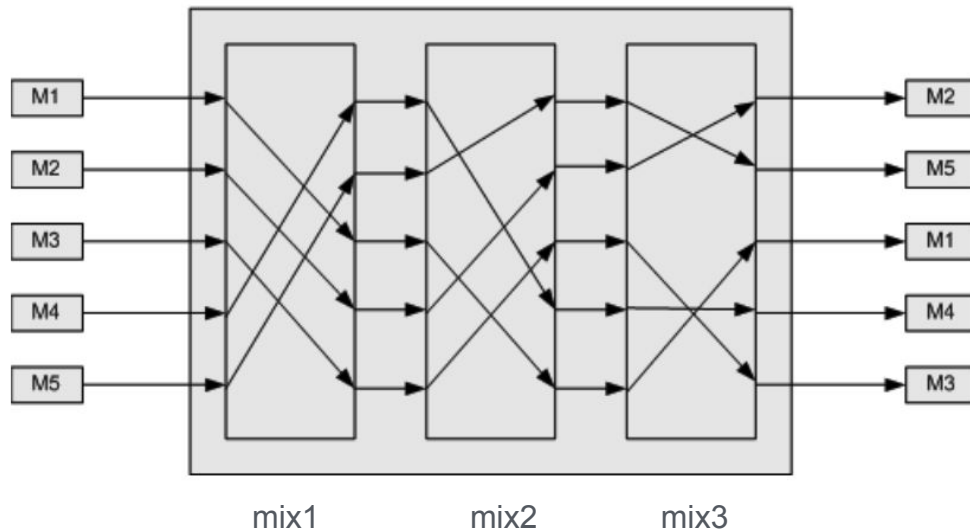
Unlinkability: cannot correlate input/output based on:

- Message contents or size.
- I/O order and timing



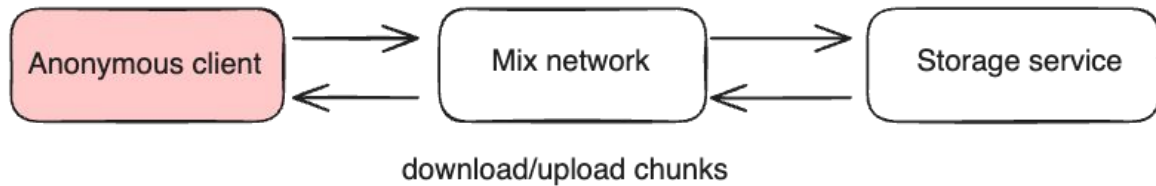
Mix protocol

- Threat model: Global Passive Adversary (GPA) observes traffic globally.
- Multiple mixes with layered encryption.
- Small packets taking random routes through the network.

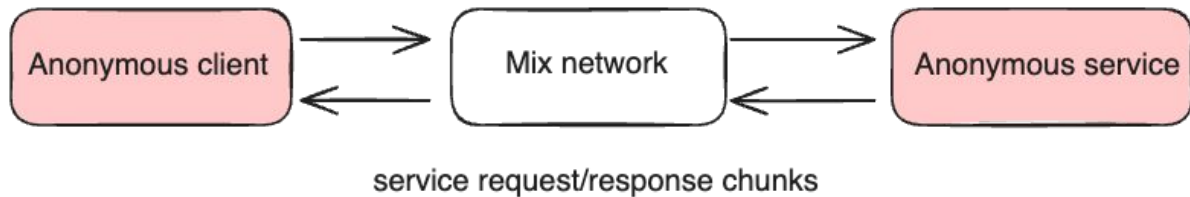


Motivation

- Anonymous download/upload



- Hidden services

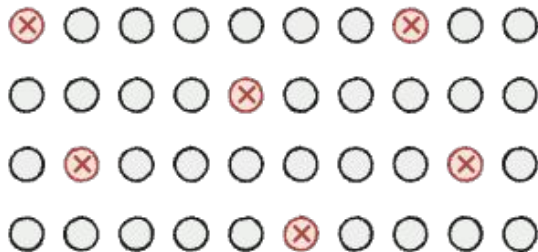


The problem: path compromise

- A file transfer require many paths.
- A hidden service repeats this exposure across many clients.

Mix Adversary: Adversary control a fraction β of the mix pool.

One pool, many independent path draws



Draw #1



Draw #2



Draw ...



Repeat until a fully malicious path occurs

Measuring De-anonymization risk

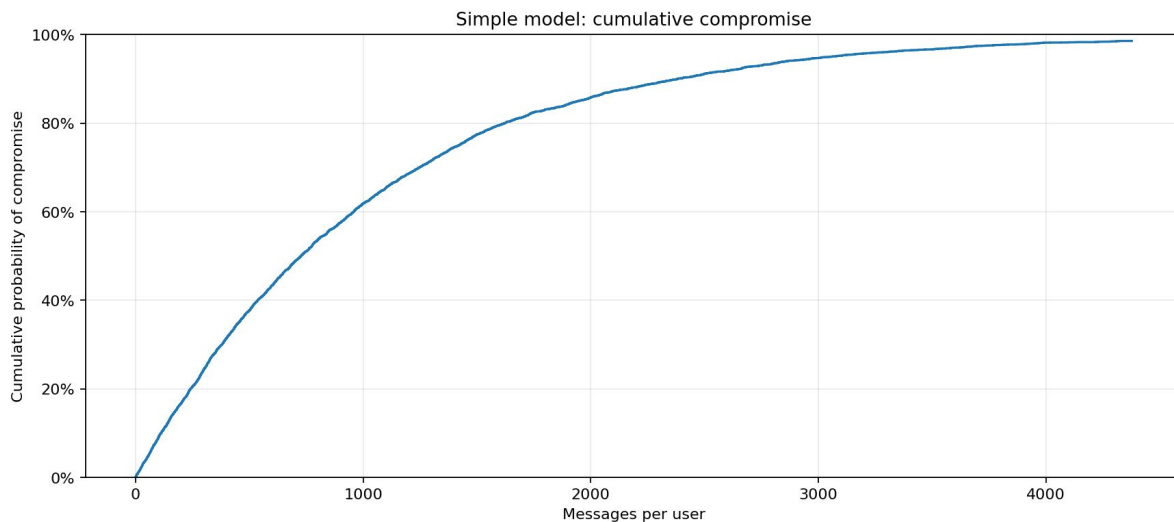
P-DLM: one fully malicious packet path

Let: β malicious %, L hops

$$q \approx \beta^L$$

S-DLM: at least one malicious path
in N selections:

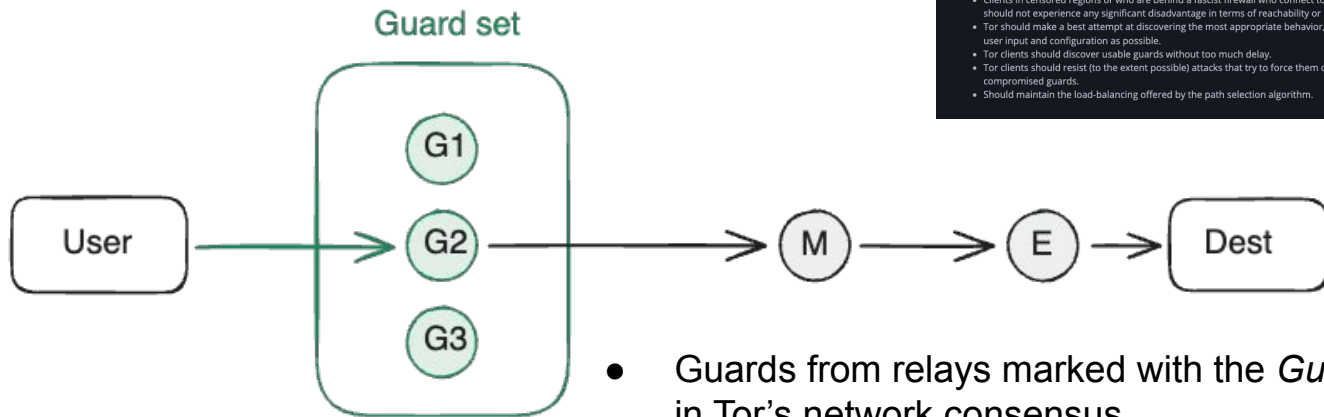
$$\text{S-DLM} = 1 - (1 - q)^N$$



RandomPathSampler | 3 hops | SybilAdversary
5,000 runs; 1000 mixes; 100 malicious (10.0%); deadline 30 days
message interval: uniform [300, 900] seconds

Existing solution: Tor guards*

- A small persistent entry set limits repeated exposure.
- Algorithms for determining how and which guards are chosen.
- Tor clients should resist attacks that try to force them onto compromised guards.
- Load balancing.



- Guards from relays marked with the *Guard* flag in Tor's network consensus.
- Use bandwidth-weighted random selection.
- Prefer primary guard.

Tor Specifications

Tor Guard Specification

Introduction and motivation

Tor uses entry guards to prevent an attacker who controls some fraction of the network from observing a fraction of every user's traffic. If users chose their entries and exits uniformly at random from the list of servers every time they build a circuit, then an adversary who had (k/N) of the network would deanonymize $F=k(N/k)^2$ of all circuits... and after a given user had built C circuits, the attacker would see them at least once with probability $1-(1-F)^C$. With large C , the attacker would get a sample of every user's traffic with probability 1.

To prevent this from happening, Tor clients choose a small number of guard nodes (e.g. 3). These guard nodes are the only nodes that the client will connect to directly. If they are not compromised, the user's paths are not compromised.

This specification outlines Tor's guard housekeeping algorithm, which tries to meet the following goals:

- Heuristics and algorithms for determining how and which guards are chosen should be kept as simple and easy to understand as possible.
- Clients in censored regions or who are behind a fascist firewall who connect to the Tor network should not experience any significant disadvantage in terms of reachability or usability.
- Tor should make a best attempt at discovering the most appropriate behavior, with as little user input and configuration as possible.
- Tor clients should discover usable guards without too much delay.
- Tor clients should resist (to the extent possible) attacks that try to force them onto compromised guards.
- Should maintain the load-balancing offered by the path selection algorithm.

*Tor guard spec: <https://spec.torproject.org/guard-spec/index.html>



Stopping Silent Sneaks: Defending against Malicious Mixes with Topological Engineering

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ABSTRACT

Mixnets provide strong meta-data privacy and recent academic research and industrial projects have made strides in making them more secure, performant, and scalable. In this paper, we focus our work on stratified Mixnets, a popular design with real-world adoption. We identify and measure significant impacts of practical aspects such as relay sampling and topology placement, network churn, and risks due to real-world usage patterns. We show that, due to the lack of incorporating these aspects in design decisions, Mixnets of this type are far more susceptible to user deanonymization than expected. In order to reason about and resolve these issues, we model Mixnets as a three-stage ‘Sample-Placement-Forward’ pipeline and develop tools to analyze and evaluate design decisions. To address the identified gaps and weaknesses we propose Bow-Tie, a design that mitigates user deanonymization through a novel adaptation of Tor’s guard design with an engineered guard layer and client guard-logic for stratified mixnets. We show that Bow-Tie has significantly higher user anonymity in the dynamic setting, where the Mixnet is used over a period of time, and is no worse in the static setting, where the user only sends a single message. We show the necessity of both the guard layer and client guard-logic in tandem as well as their individual effect when incorporated into other reference designs. We develop and implement two tools, 1) a mixnet topology generator (Mixnet-Topology-Generator (MTG)) and 2) a path simulator and security evaluator (routesim) that takes into account temporal dynamics and user behavior, to assist our analysis and empirical data collection. These tools are designed to help Mixnet designers assess the security and performance impact of their design decisions.

CCS CONCEPTS
• Security and privacy → Network security; Pseudonymity, anonymity and untraceability.

ACM Reference Format:

Xinshu Ma, Florentin Rochet, and Tariq Elahi. 2022. Stopping Silent Sneaks: Defending against Malicious Mixes with Topological Engineering. In *Annual Computer Security Applications Conference (ACSAC ’22)*, December 5–8, 2022, Austin, TX, USA. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3564625.3567996>

1 INTRODUCTION

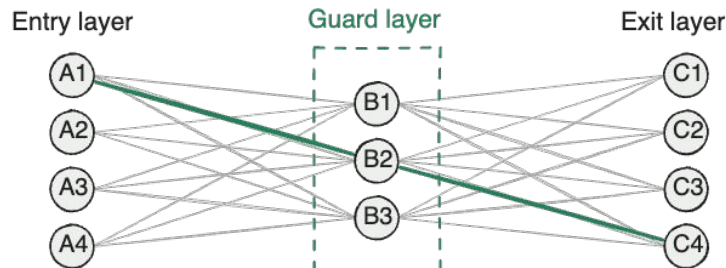
Since the ‘Five Eyes’ mass surveillance disclosures by Snowden high-lighted real-world adversaries’ pervasive and global nature, we observe a greater community focus on strong meta-data privacy to protect and improve communication protocols on the Internet [26]. Tor, with ≈ 8 million daily users [43], provides limited protection against a global adversary and traffic analysis attacks [35, 47, 55]. Thus, there is a resurgent interest in mix networks (Mixnets) [8]—once considered impractical to deploy—with many recent proposals from academia [7, 36, 38, 39, 51, 68, 70] and industry [18] that have strong security guarantees and improved performance at scale.

The security of many known designs, such as Vuvuzela [70], Karaoke [39], Loopix [51], and Nym [17] rely on the *anystrat* assumption where at least one server in the user’s path must be honest. In other words, security comes from distributing trust across many relay operators. Practical real-world designs distribute trust and provision network resources by drawing from third-parties, such as volunteers or for-profit participants, on which the network applies light (e.g., Tor’s path selection IP restrictions) or no constraints. These third-parties can be malicious and it is critical that the mixnet design resist their influence. In Mixnet literature, the security analysis typically considers active attacks like traffic analysis [1], (n-1) [39], and Denial-of-Service (DoS) [5]. In addition, users can also be deanonymised by passive adversaries whenever a message traverses a path composed entirely of adversarial relays.

However, the literature typically takes for granted real-world issues such as network configuration and routing, network churn,

Existing solution: Bow-Tie*

- **Adapting guards to stratified mixnets:** guard layer with client logic that reuses guards and limits exposure to new ones.
- **Accounting for churn:** maintaining stable guards to reduce replacement-driven exposure.
- Stratified network topology: balancing bandwidth across layers.
- **Evaluating risk over time:** studying cumulative deanonymization and time to first compromise using their **routesim** simulator.



*Stopping Silent Sneaks: Defending against Malicious Mixes with Topological Engineering:

<https://dl.acm.org/doi/pdf/10.1145/3564625.3567996>

Existing solution: When Mixnets Fail, NDSS 2026*

- Considers **sessions**: a set of related packets belonging to one client-destination communication.
- Demonstrate that exchanging **4MB results in full de-anonymization** of the client. ($\beta = 10\%$)
- Propose **path selection strategies** for sessions: K-HF, K/W, α -sticky
- Introduce new **evaluation metrics** to measure the probability of de-anonymization (*DLM*).
- Provide **simulator** to confirm evaluation results.

When Mixnets Fail: Evaluating, Quantifying, and Mitigating the Impact of Adversarial Nodes in Mix Networks

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Abstract—Mix networks (mixnets) provide clients with communication anonymity against strong network adversaries by traversing their packets independently through randomly selected hops (mixnodes), which disrupt packet linkability. Although this approach, implemented in Nym, maximizes obfuscation against network adversaries, it enables an adversary who compromises a subset of mixnodes (10%/5% of nodes) to entirely nullify the anonymity of all clients whose communication volume with their destination exceeds a certain threshold (4 MB/30 MB). To mitigate such vulnerabilities, this work develops a set of novel path selection techniques that achieve a trade-off between resistance to network adversaries and resilience against compromised mixnodes. Observing that existing anonymity metrics are insufficient to quantify adversarial risk in mixnets, we additionally introduce effective empirical and simulation-based metrics. Through theoretical, empirical, and simulation-based evaluations, we comprehensively assess our proposals, demonstrating that the proposed approaches reduce the vulnerability to compromised nodes by up to 80%, while conferring limited advantage to network adversaries. Our analysis further reveals that state-of-the-art anonymity metrics, in contrast to our proposed metrics, produce misleading results that influenced certain design choices in Nym.

I. INTRODUCTION

Concealing the relationship between communicating parties on the Internet, mix networks (*mixnets*) [6], [15], [5], [41], [25], [21], [9] stand out as an effective anonymous communication system. Mixnets operate as overlay networks that forward clients' communication packets through multiple intermediaries (*mixnodes*), randomizing their input traffic order to render tracing packets back to their originators (clients) infeasible for a network adversary observing all communication exchanges, commonly referred to as a *Global Passive*

currently deployed in practice—the Nym network [9]. In this design, nodes are arranged into layers such that each node in layer l is only connected to nodes in layers $l-1$ and $l+1$. Consequently, each packet traverses exactly one node from each layer before reaching its destination. Additionally, for mixing purposes in Loopix, each mixnode flushes incoming packets after a random delay sampled from an exponential distribution [18], a suitable scheme for real-time communications.

Within Loopix (Nym), client traffic is transmitted to its destination by fragmenting the traffic data into fixed-size packets using the Sphinx packet format [8], an efficient scheme that enhances packet unlinkability [25], [9]. Notably, each Sphinx packet carries a small payload (2 KB in Nym¹) and is routed independently via a path selected uniformly at random, consisting of one node from each mixnet layer. This design ensures that each route consistently carries a comparable number of packets, thereby reducing the GPA's ability to infer communication patterns based on traffic volume [9].

Although the aforementioned design choice effectively thwarts the threat posed by the GPA, its impact on the advantage of *mixnode adversaries*, compromising a subset of nodes within the mixnet, has not been thoroughly examined. Specifically, such adversaries can fully deanonymize clients if any of their packets traverse only adversarial nodes. While the threat of mixnode adversaries has been largely overlooked in state-of-the-art mixnet works [9], [16], [34], [35], their lower infrastructure requirements make them a more practical threat in real-world scenarios.

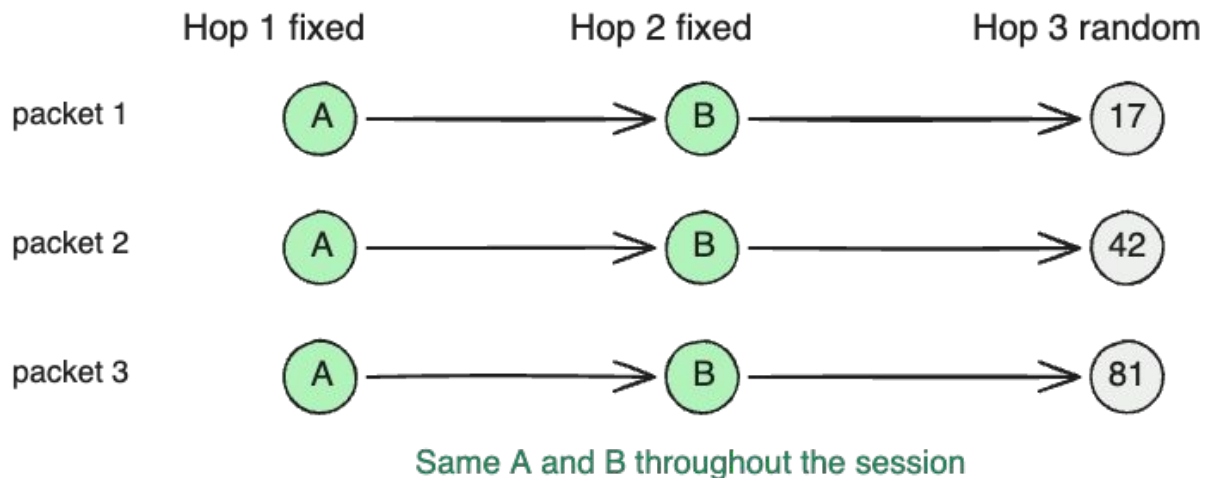
Design Goals. Given the practical threat of mixnode adver-

*When Mixnets Fail: Evaluating, Quantifying, and Mitigating the Impact of Adversarial Nodes in Mix Networks: <https://www.ndss-symposium.org/wp-content/uploads/2026-f2384-paper.pdf>

Session strategy: K-Hops Fixed (**K-HF**)

- Fix h hop positions for the session.
- Any honest fixed node blocks full-path compromise.

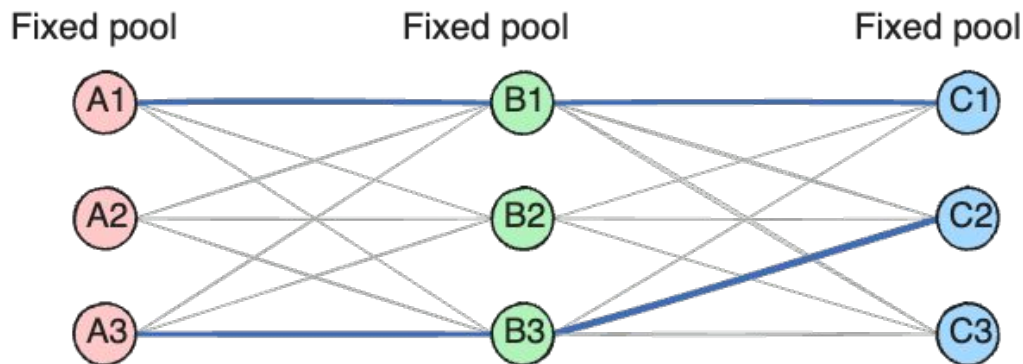
$$S\text{-DLM}_{\text{K-HF}}(N) \approx \beta^{h_f} \cdot \left(1 - (1 - \beta^{L-h_f})^N\right)$$



Session strategy: **K/W**

- Persistent candidate pools offer more choices than one fixed node.
- K controls exposure and path diversity.

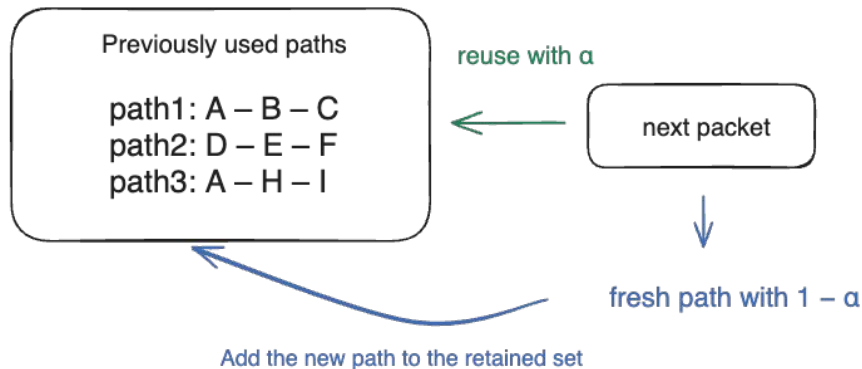
$$S\text{-DLM}_{K/W}(N) \lesssim 1 - (1 - \beta^L)^{\min(N, K^L)}$$



Session strategy: α sticky selection (**α -sticky**)

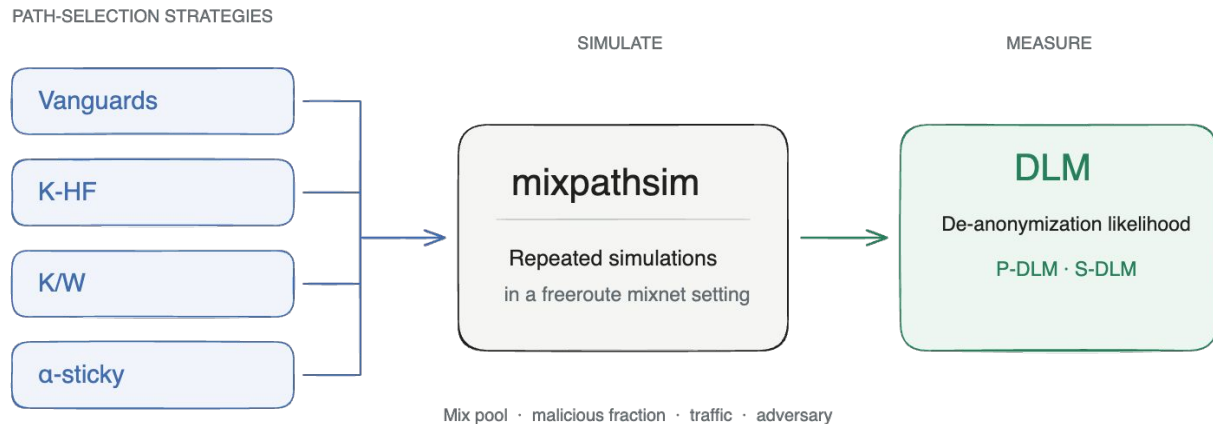
- Sample one initial path and keep it in a set. For each later message, reuse a path from the set with probability α , or introduce a new path with probability $1-\alpha$.
- Larger α slows fresh exposure. For $\alpha < 1$, new paths continue to enter the retained set.

$$S\text{-DLM}_{\alpha\text{-SS}}(N) \approx 1 - (1 - q)(1 - (1 - \alpha)q)^{N-1}$$



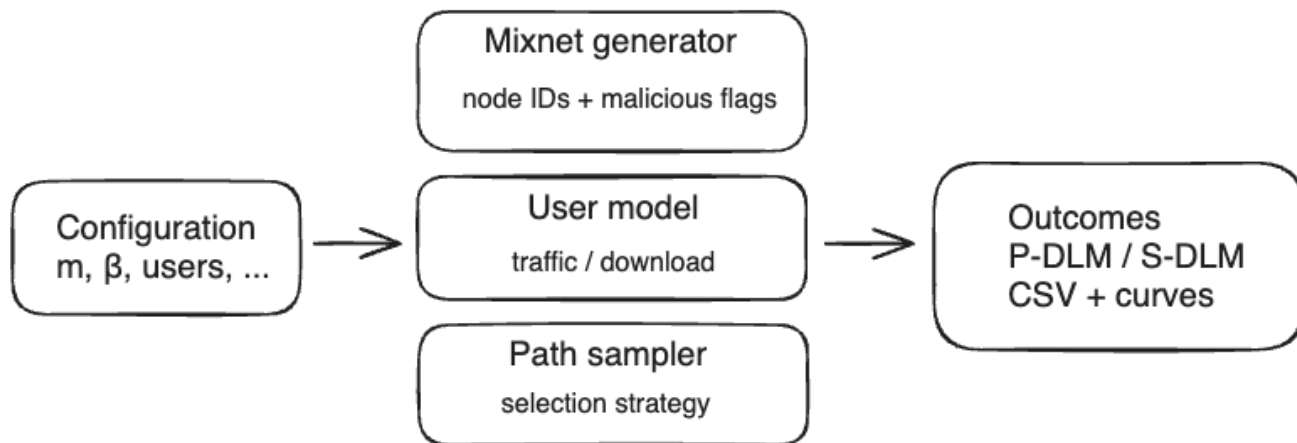
Finding the optimal session strategy

- Simulate in freeroute mix setting
- Repeat simulation for different configurations of the strategies.



Simulator: **mixpathsim***

Monte Carlo trials estimate compromise likelihood across users with independent selection state.

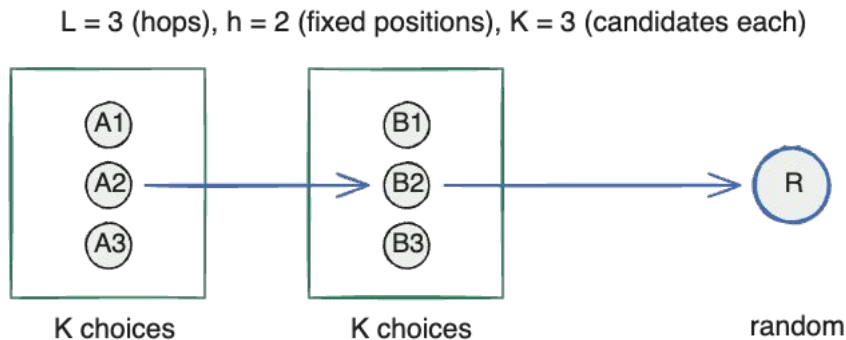


*mixpathsim: <https://github.com/logos-storage/hs-mix-sim>

A hybrid strategy: K candidates at h fixed hops

Generalize K-HF and K/W:

- L hops, h persistent positions with K candidates each and $L-h$ random positions.
- This is a configurable trade-off: smaller K reduce exposure but provide less path diversity.
- Tune L , h and K for the application.



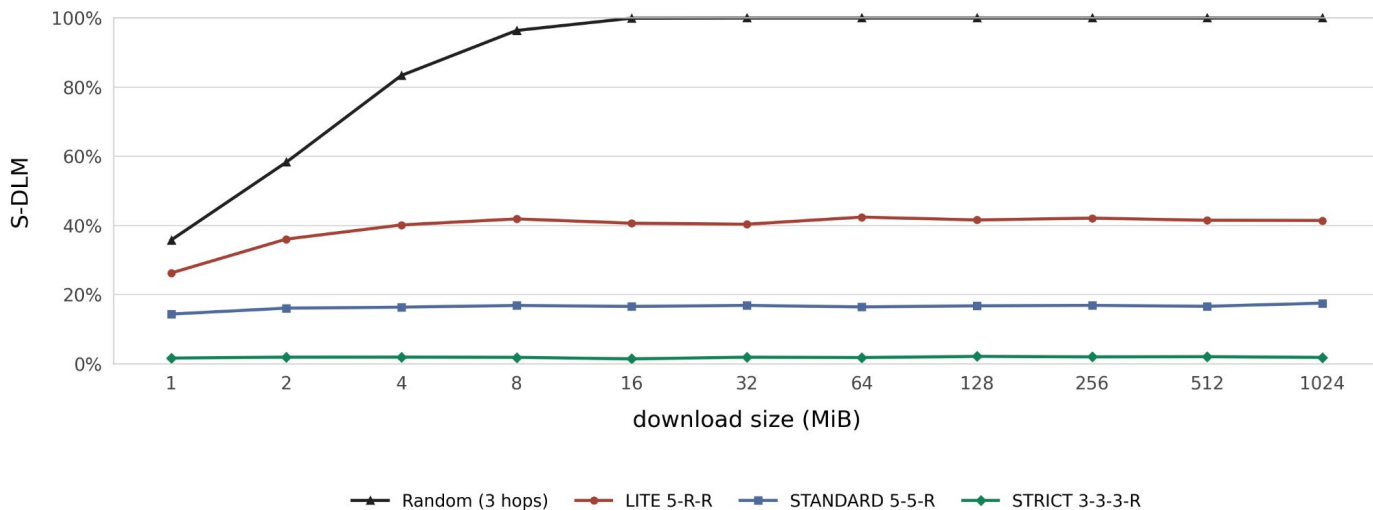
A hybrid strategy: session profiles

- Select a small pool of candidates for each fixed position at the start of a session.
- For each packet, choose one node from each retained pool; choose R freshly from the general mix pool.
- Retain these pools throughout the session, then select new pools for the next session.

Profile	Path layout	Configuration
LITE	5 → R → R	One fixed pool of five nodes; two random hops
STANDARD	5 → 5 → R	Two fixed pools of five nodes; one random hop
STRICT	3 → 3 → 3 → R	Three fixed pools of three nodes; one random hop

A hybrid strategy: session experiments.

Results measure path-compromise likelihood for different configurations depending on the required anonymity level and path diversity.

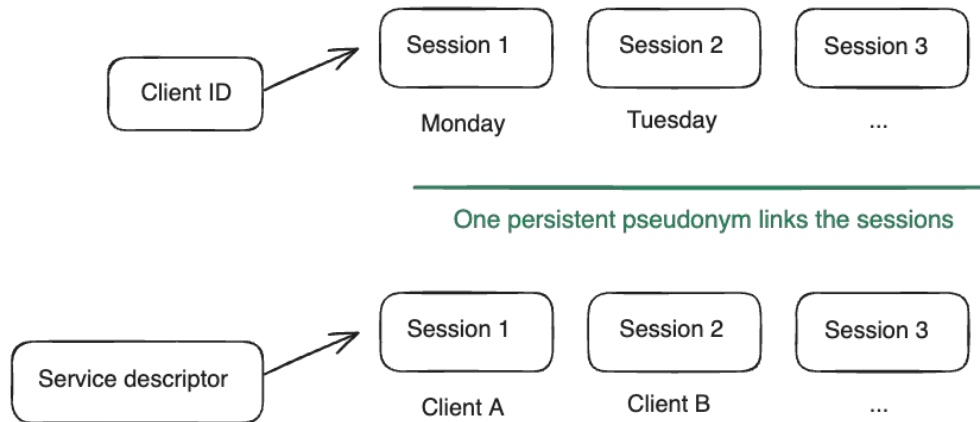


Demo

Problem: persistent pseudonym

Persistent pseudonyms outlive sessions, example use-cases:

- Public provider: persistent client ID. i.e., **anonymous** clients of **public** services/storage providers
- Hidden service: stable service ID (descriptor) shared across many clients. i.e., **anonymous** clients of **anonymous** services.



Can session strategies protect a long-lived communication over mix?

Options:

1. New session state.

Problem: increase the chance of choosing malicious fixed nodes over time.

2. Reusing the same state

Problem: the same state gives the *adaptive* adversary persistent targets.

Adaptive adversary: discovers and targets additional nodes over time.

Path walking: work back from the exit toward the service

Each controlled node reveals the next node toward the service.

Service traffic →

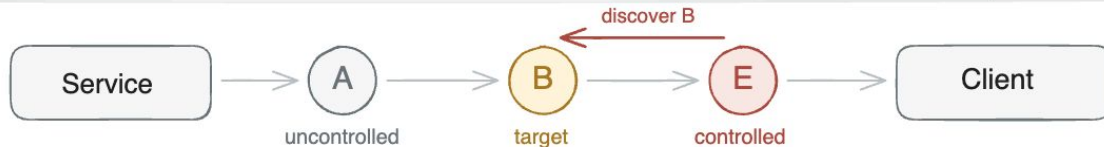
□ Uncontrolled

■ Targeted

■ Controlled

1 Start at a controlled exit

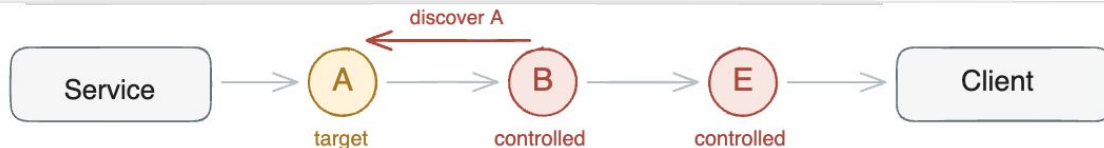
E reveals B.
The adversary targets B.



Time to compromise B

2 Gain control of B

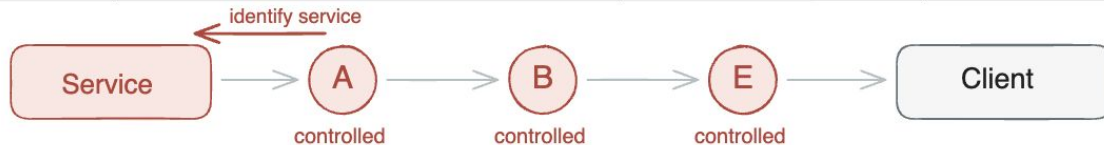
B reveals A.
The adversary targets A.



Time to compromise A

3 Gain control of A

The controlled chain now reaches the service.

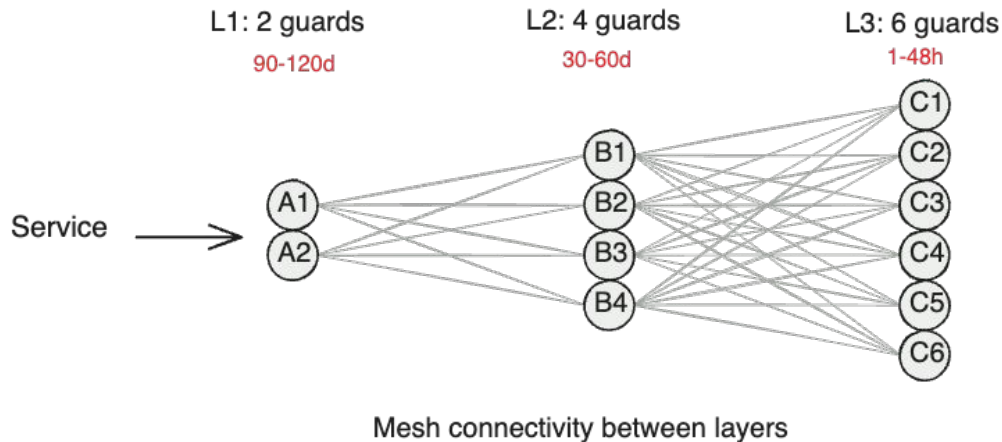


Assume still-active route. Compromise attempts take time and may fail.

Time-based selection strategies

Starting point: **Tor Vanguards**

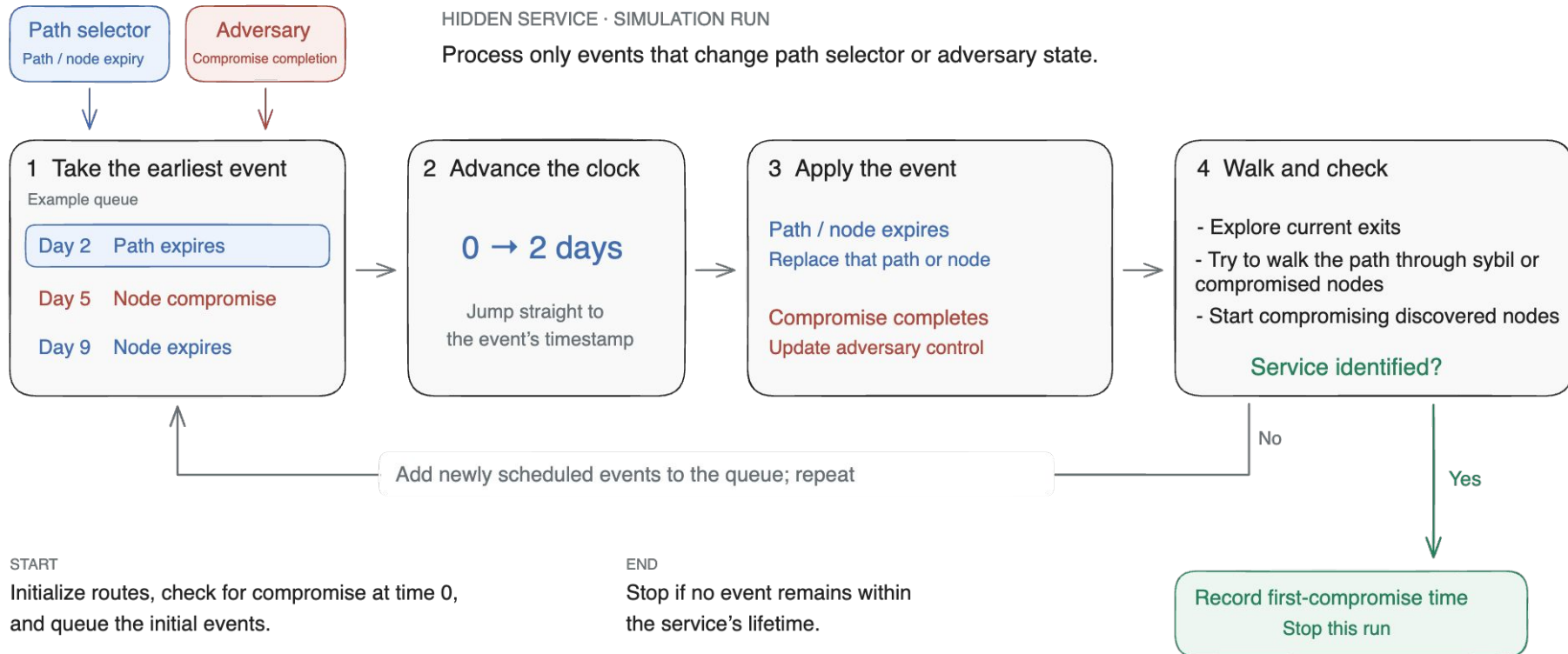
- Layers of guard nodes, each layer with its own lifetime sampled from a defined range.
- Layers closer to service have higher lifetimes than ones further away.
- “Incentivize” the attacker to sybil L3 and then compromise L2 and L1.



Does a Vanguard-inspired strategy work over mix?

- A free-route mix has different path and traffic assumptions.
- The comparison needs an event model and an adversary that can walk current routes.
- We need to extend our simulator to support time-based path selection and a given adversary model (the adaptive adversary).

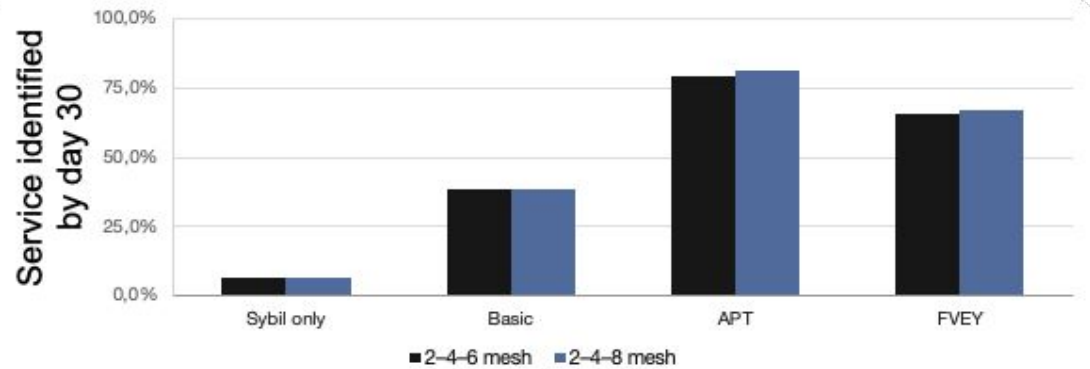
Discrete event simulation



Vanguard baseline under the mix model

Simulation config:

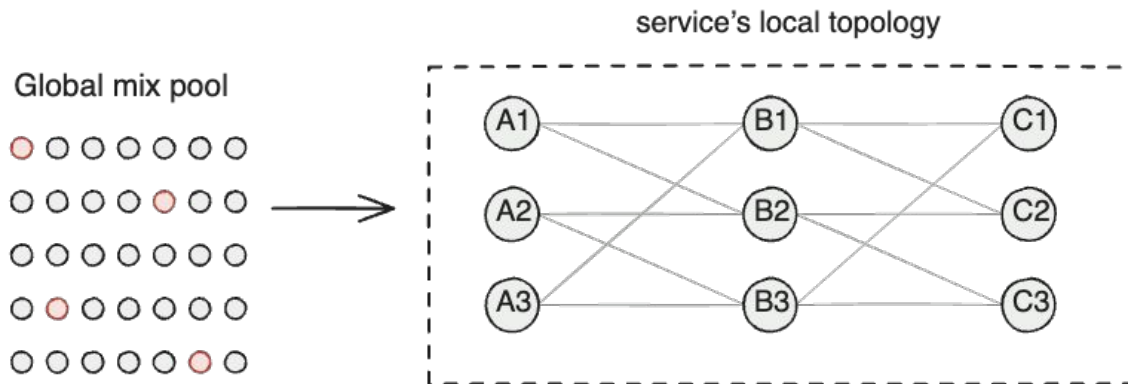
- 5,000 trials,
- 1,000 nodes,
- $\beta = 10\%$,
- 30 days.



Adversary	Success and delay after targeting
Sybil only	Initial malicious nodes, no extra compromises
Basic	50% within 15 days, otherwise never
APT	75% within 15 days, 100% by 30 days
FVEY	50% within 2 days, 75% by 7 days, else never

Can we do better?

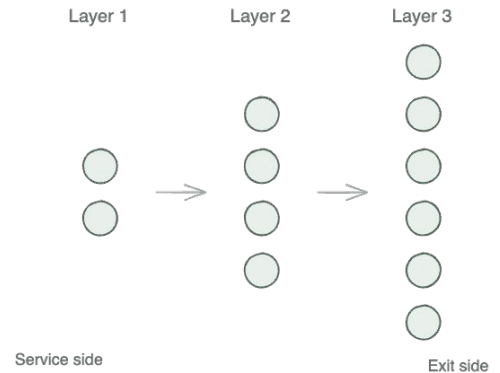
- First step: Generalize the vanguard approach to support any configured params:
 - Local topology: a private layered graph with the user/service selection over the global free-route mix pool.
 - A topology describes candidate nodes at each hop (K), generalizes the mesh and allow any degree of connections (d) between layers, and independent node and path lifetimes (T).
- Second step: Simulate/experiment ...



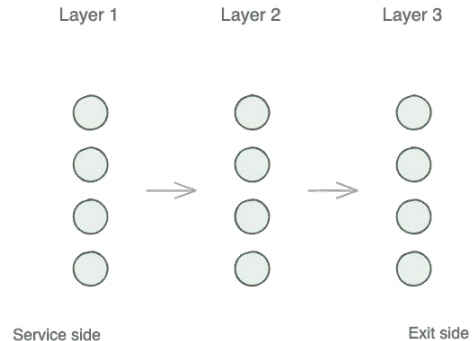
Observation 1: Rethinking layer sizes for mixnets

- Vanguards motivates protecting a small set of service-side guards against discovery.
- In our mixnet compromise model, an honest node anywhere along the path prevents full-path compromise.
- We therefore explore alternatives: equal-sized layers, different distributions of nodes, and independent rotation times.
- Equal-sized layers: the same total of 12 nodes, distributed equally.
- We evaluate these choices together against an adaptive adversary.

Vanguard layers



Equal-sized layers

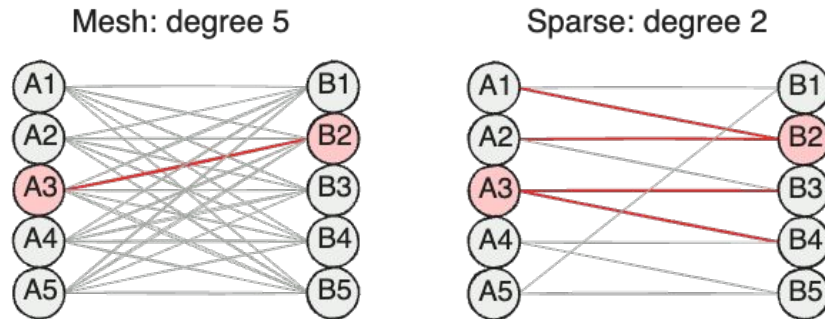


Observation 2: Degree controls connectivity and exposure

- Increasing degree creates more possible routes and more alternatives when a node becomes unavailable.
- Lowering degree restricts which nodes can form a complete path.

Main point:

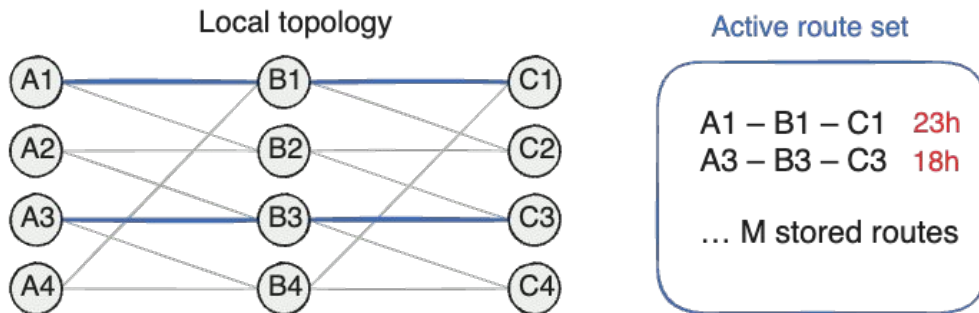
- Malicious nodes in every layer are insufficient for compromise unless they have a connected route, therefore, a sparse topology can exclude malicious combinations that exists in a full mesh.



Observation 3: Active paths limit exposure

- A local topology defines the routes that are possible.
- We select only M complete routes for active use.
- Each message chooses from this active set; each route has its own lifetime.
- Expiring routes change the combinations available to the adversary while retaining the underlying topology.

Main point: We can maintain a larger candidate topology while exposing only a small set of complete routes at any moment.

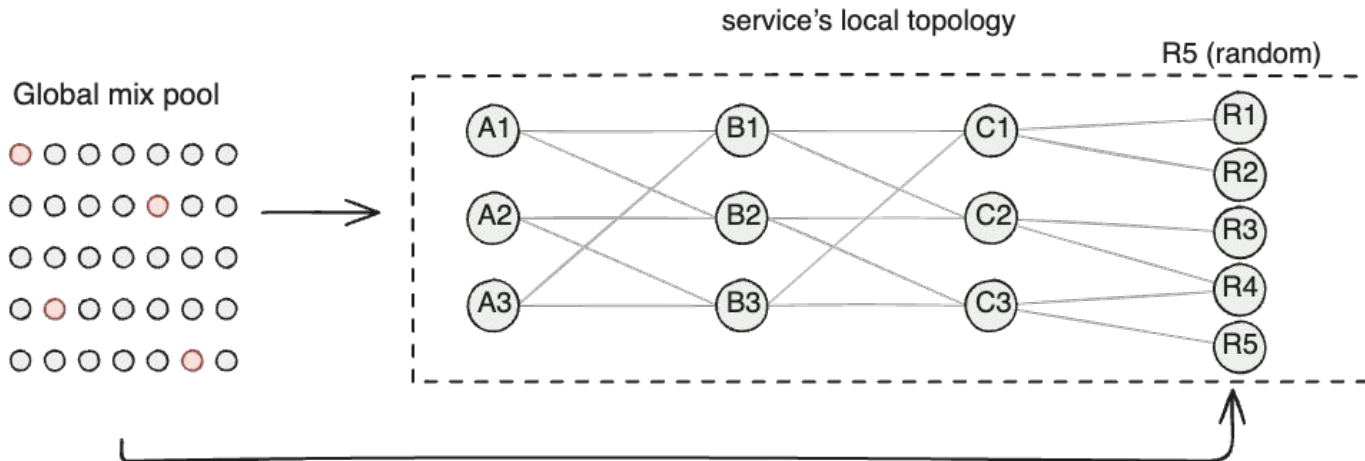


Observation 4: Rotating exit layer

- Add a client-facing layer whose nodes come from the general mix pool.
- Rotate these nodes independently of the internal layers and active paths.
- A node may leave the topology before a targeted compromise completes.

Main point:

- Exit short rotation limits the time available to compromise an exit, therefore, “incentivize” sybil and delay compromise of internal nodes
- Rotating exit can delay useful access to internal compromised nodes.



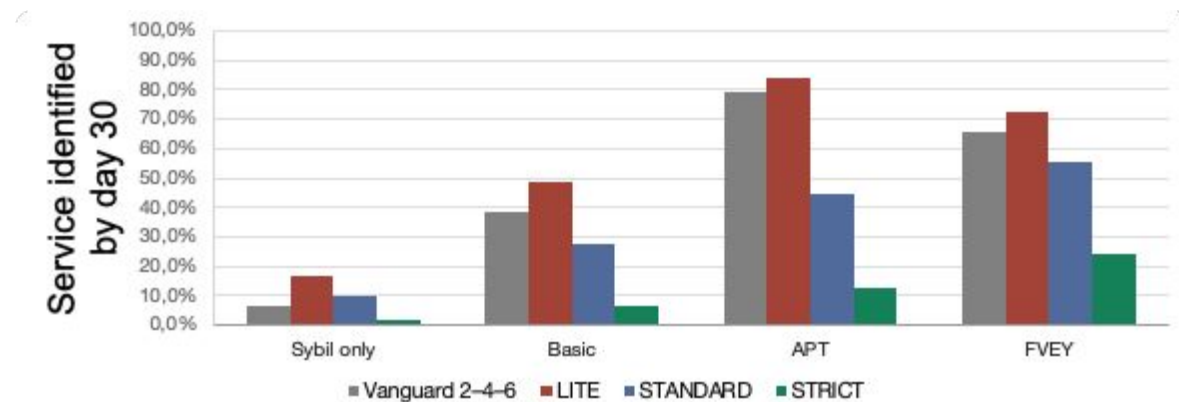
Simulation results: Time-based anonymity profiles

Profile	Topology	Degree	Active routes	Median time to service identification			
				Sybil only	Basic	APT	FVEY
LITE	5-5-R	Mesh	5 prefixes	NR	NR	14.39 d	5.64 d
STANDARD	5-5-R5	3	5 routes	NR	NR	NR	24.04 d
STRICT	5-5-5-R5	3	5 routes	NR	NR	NR	NR

NR: fewer than 50% of services were identified within 30 days; median not reached.

5,000 runs per case · 10% initially malicious

R: fresh random hop per message, R5: five candidate slots, each with its own rotation timer



Demo

Limitations

- **SURBs and mix key rotation:** Rotating active paths does not invalidate old SURBs. The routing-key extend exposure beyond the modeled path lifetime.
- **Assumed adversary capabilities:** Results depend on adversary time to compromise, success probabilities and number of attacks.
- **Trusted-pool assumption:** Lower malicious percentage is assumed.
- **Path discovery:** The adversary is assumed to be able to launch unlimited traffic volume to the service.
- **Side channel attacks:** We measure full-path compromise, not all forms of deanonymization, such as traffic analysis or side channel attacks.
- **Churn and mix node failures:** we assume mostly stable nodes at least in the trusted-pool.
- **Evaluation period:** we consider 30 days which doesn't establish safety over longer lifetimes.

Takeaways

- **Repeated random selection accumulates risk.** More packets create more opportunities to select a fully malicious path.
- **Session strategies limit exposure.** Fixed pools balance compromise risk and path diversity within a session.
- **Persistent identities need protection across time.**
- **Time-based profiles combine topology, active-path limits and rotation.** Lowers the probability of de-anonymization, but limits the time a service is available.
- **Protection depends on the configuration and threat model:** Pool quality, attacker capabilities, and availability requirements.

Thank you!

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<https://github.com/logos-storage/hs-mix-sim>